Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Met Office

Global Ocean OSTIA Near Real Time Level 4 Sea Surface Temperature Product

SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001

Issue: 6.0

Contributors: S. Good, C. Mao.

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CHANGE RECORD

When the quality of the products changes, the QuID is updated, and a row is added to this table.

The third column specifies which sections, or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	15/12/2014	all	First version of document	A. McLaren et al.	
1.1	26/02/2015	I, II	Revision after V5 acceptance	A. McLaren et al.	
1.2	May 1 2015	all	Change format to fit CMEMS graphical rules		L. Crosnier
1.3	February 23 2016	II.4, V	Update for new data sources	E. Fiedler	B. Hackett
1.4	September 20 2016	II.4, II.5.2, V	Update reference instrument.	S. Good, C. Mao	
1.5	September 19 2017		Change data assimilation scheme to NEMOVAR.	S. Good	
1.6	January 17 2019	II.3, II.4, II.5, V I.2, III, doc ref	Improvement to feature resolution and use SLSTR-A SSTs. Update references to CMEMS elements.	S. Good	
1.7	March 22 2019	all	Review of the document	S. Good	Mercator Ocean
1.8	September 6 2019	II.4, VI	Upstream data change	S. Good	
1.9	September 10 2020	all	Change to reference dataset used for bias correcting satellite SSTs and other minor changes and formatting improvements	S. Good	
3.0	16 June 2022	V, VI	Upstream data change.	S. Good	A. Pisano
4.0	15 December 2022	I.1	Handover of activities due to Brexit	D. Briand	A. Pisano
5.0	24 January 2025	all	Updates for temporal extent of the dataset, removal of some datasets and general review	S. Good	A. Pisano
6.0	24 June 2025			S. Good, C. Mao	A. Pisano

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
EAN	Estimated Accuracy Number
GDS	GHRSST Data Specification
GHRSST	Group of High-Resolution Sea Surface Temperature
MY	Multi-year
NRT	Near Real Time
OSTIA	Operational SST and Ice Analysis
REP	Reprocessing
SST	Sea Surface Temperature
TAC	Thematic Assembly Centre
UK	United Kingdom

KEY DEFINITIONS

Foundation sea surface temperature	**CF Definition**: The surface called "surface" means the lower boundary of the atmosphere. The sea surface foundation temperature is the water temperature that is not influenced by a thermally stratified layer of diurnal temperature variability (either by daytime warming or nocturnal cooling). The foundation temperature is named to indicate that it is the temperature from which the growth of the diurnal thermocline develops each day, noting that on some occasions with a deep mixed layer, there is no clear foundation temperature in the surface layer. In general, sea surface foundation temperature will be similar to a night-time minimum or pre-dawn value at depths of between approximately 1 and 5 meters. In the absence of any diurnal signal, the foundation temperature is considered equivalent to the quantity with standard name sea_surface_subskin_temperature. The sea surface foundation temperature defines a level in the upper water column that varies in depth, space, and time depending on the local balance between thermal stratification and turbulent energy and is expected to change slowly over the course of a day. If possible, a data variable with the standard name sea_surface_foundation_temperature should be used with a scalar vertical coordinate variable to specify the depth of the foundation level. Sea surface foundation temperature is measured at the base of the diurnal thermocline or as close to the water surface as possible in the absence of thermal stratification. Only in situ contact thermometry is able to measure the sea surface foundation temperature. Analysis procedures must be used to estimate sea surface foundation

	temperature value from radiometric satellite measurements of the quantities with standard names sea_surface_skin_temperature and sea_surface_subskin_temperature. Sea surface foundation temperature provides a connection with the historical concept of a "bulk" sea surface temperature considered representative of the oceanic mixed layer temperature that is typically represented by any sea temperature measurement within the upper ocean over a depth range of 1 to approximately 20 meters. The general term, "bulk" sea surface temperature, has the standard name sea_surface_temperature with no associated vertical coordinate axis. Sea surface foundation temperature provides a more precise, well-defined quantity than "bulk" sea surface temperature and, consequently, is more representative of the mixed layer temperature. The temperature of sea water at a particular depth (other than the foundation level) should be reported using the standard name sea_water_temperature and, wherever possible, supplying a vertical coordinate axis or scalar coordinate variable.

RELEVANT DOCUMENT LIST

Product User Manual	https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-011.pdf
Synthesis Quality Overview Document	https://documentation.marine.copernicus.eu/SQO/CMEMS-SST-SQO-010-011.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the global ocean near real time SST product **(SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001).** This is a level 4 (L4) daily analysis product on a 0.05° grid, produced using the Operational SST and Ice Analysis (OSTIA) system (Donlon et al., 2012) at the UK Met Office.

Table 1. Details of the global ocean near real time SST product (SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001).

Time period	-2Y - Present
Spatial resolution	0.05°
Minimum to maximum Longitude	-180° to 180°
Minimum to maximum Latitude	-90° to 90°
Daily fields in principle dataset	Analysed SST, sea ice area fraction, mask, SST analysis error

I.2 Summary of the results

The accuracy of the SST analysis product has been assessed using a comparison to drifting buoy and near surface (3-5 m) Argo data. Drifting buoy data is used in producing the SST product and therefore is not an independent dataset. However, only a subset of the available drifter data is used for validation. Argo data is independent of this product, but has a reduced coverage compared to drifting buoys. The drifting buoy data assessment was carried out for the year 2012; however, a recent Argo assessment is available for June 2020.

The mean difference relative to drifter data in 2012 was very small, with a global mean difference of 0.06 K and regional mean differences of less than 0.1 K except in the Mediterranean and Arctic regions which suffer from a lack of available observations. The standard deviation of differences was 0.40 K globally with regional values ranging from 0.28 K in the South Pacific to 0.89 K in the Mediterranean.

Similarly, in the most recent assessment that compared to Argo data, the mean difference relative to Argo data was very small, with a global mean difference of -0.04 K and regional mean differences ranging from 0.04 K in the North Atlantic to -0.09 K in the Indian and Southern Oceans. The standard deviation of Argo differences is 0.34 K globally, with lowest values of 0.18 K in the Tropical Pacific and highest value of 0.48 K in the North Atlantic.

The independent Argo observations from 2013 were also used to assess the analysis error estimates. The standard deviation of the differences to Argo were compared to the analysis error estimates at the matchup locations. Results suggest that the analysis error is underestimated in all regions compared to Argo. However, this is contrary to a separate study which found that OSTIA reanalysis error estimates compared well with drifting buoy differences (Corlett et al., 2014) as discussed further in section 0. Further work is required to examine the difference in the methodologies for assessing the error estimates before a firm conclusion on the accuracy of the analysis error estimates can be provided.

I.3 Estimated Accuracy Numbers

Global estimated accuracy numbers are provided for June 2020 based on a comparison with Argo data.

Table 2. Global difference statistics between the SST analysis product and Argo observations for June 2020.

Variable	Units	Metrics	Values of metric
temperature	K	Mean difference	-0.04
temperature	K	Root Mean Square (RMS)	0.34

II PODUCTION SYSTEM DESCRIPTION

II.1 Production centre

Met Office, United Kingdom (UK)

II.2 Production subsystem

Operational SST and Ice Analysis (OSTIA) system

II.3 Product description

The level 4 product (Table 3) is a satellite and in-situ foundation SST analysis created by the OSTIA (Operational SST and Ice Analysis) system. The product is created daily on a global regular grid at 0.05° resolution and data is available for the previous two years up to the present time. The OSTIA product provides an estimate of the foundation SST which is the SST free of diurnal variability. In addition to the foundation SST, the daily product contains sea ice fraction, land/sea ice/lake mask and an estimate of the SST analysis error.

Table 3. Description of SST_GLO_SST_L4_NRT_OBSERVATION_010_001.

Product Specification	SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001 (OSTIA)
Short Description	L4 daily gap-free maps of foundation SST on a global regular grid with daily satellite bias and anomaly maps, and monthly and seasonal SST averages
Variables/datasets	METOFFICE-GLO-SST-L4-NRT-OBS-SST-V2 – daily SST
Reprocessing	No
Geographical coverage	Global
Spatial resolution	0.05 x 0.05 degree
Vertical coverage	Foundation SST
Temporal coverage	-2Y – present
Temporal resolution	Daily
Input satellites / sensors	Various including SNPP VIIRS, Sentinel 3A SLSTR, MetOp B AVHRR, GCOM-W1 AMSR2, MSG SEVIRI, SSMIS, MetOp-C AVHRR.
Target delivery time	Daily files: by 12:00 UTC on the following day
Delivery mechanism	Subset / files / maps
Update Frequency	Daily

II.3.1 SST Level 4 reprocessing Chain and Algorithms

The Operational Sea surface Temperature and Ice Analysis (OSTIA) system is run daily in the operational suite at the Met Office. The product has a target delivery time of 12:00 UTC but are usually available by 08:15 UTC.

Figure 1 shows the different steps for the creation of the OSTIA products at the Met Office. Each step of this processing is described below. Further information is contained in Good et al. (2020) and Donlon et al. (2012).

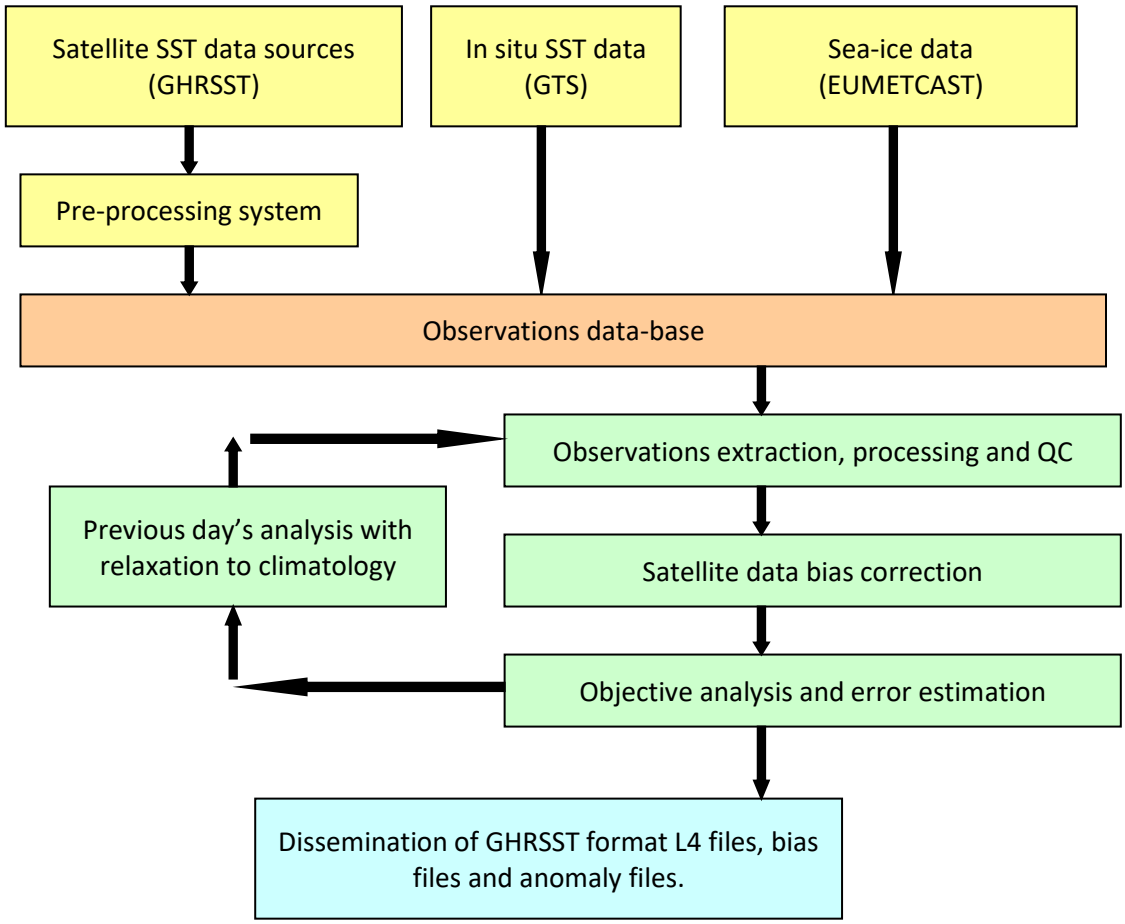

Figure 1. Schematic diagram of the operational OSTIA processing chain at the Met Office.

II.4 Collection of inputs

The following inputs are collected as inputs to OSTIA:

SST satellite data: The original data from different producers (including NASA, NOAA, IFREMER, EUMETSAT OSI-SAF, ESA) in GHRSST compliant format are collected. The format of the collected data is that recommended by the GHRSST [1]GDS (Team, J., E., K., & C., 2022).

All data are provided with Sensor Specific Error Statistics (SSES_bias and SSES_std variables) that give respectively an estimate of the systematic and random errors at pixel level. In addition, quality level flags as specified by the GHRSST are provided .

The data used in OSTIA include both infra-red and microwave.

Infra-red data:

- Data from the AVHRR instruments on board NOAA and MetOp-B and -C satellites;
- SEVIRI on board the MSG satellites;
- VIIRS data on board Suomi NPP and NOAA20; and
- SLSTR data from Sentinel 3A and Sentinel 3B.

Microwave data: AMSR2 instrument on board GCOM-W1.

In situ SST data: from ships, drifting and moored buoys are used in OSTIA.

Ice concentration data: from the EUMETSAT OSI-SAF are used in OSTIA for the ocean. Since mid-April 2013 NCEP data have also been used for ice in inland waters.

Climatologies: the SST climatology used in the OSTIA background relaxation step has been derived from the OSTIA reanalysis (SST-GLO-SST-L4-REP-OBS-010-011), produced as part of the MyOcean projects that preceded the Copernicus Marine Service.

Lake Surface Water Temperature (LSWT) data: is available within some GHRSST L2P files. These data are processed by the data providers using the SST algorithms rather than lake-specific algorithms (which take into account the emissivity and altitude of the lakes) and are therefore less accurate than the data over the open ocean. In situ data over lakes are also available over the GTS, mainly for the Great Lakes.

II.5 Algorithms for L4 production

II.5.1 Quality control and pre-processing of input data

All satellite and in situ SST data valid for a particular day, with a 6-hour overlap on the days either side (which helps reduce temporal noise in the analyses), are extracted from the Met Office observations database. Data that were assimilated in the previous cycle are not used again. The input SST data undergo various QC and processing steps:

[1] https://www.ghrsst.org/

- only satellite data which have a quality flag of 4 and higher are accepted, except for AATSR and VIIRS data where only data of quality 5 are accepted.
- a Bayesian statistical check against the background field is carried out for all input data using the same error covariance information used to perform the objective analysis later in the process.
- a diurnal check is carried out whereby day-time data (determined using a solar zenith angle calculation) with a wind-speed of less than 6m/s are rejected (see e.g. Donlon et al., 2002).
- the SSES biases supplied with the GHRSST data are removed from each pixel and the SSES standard deviation values are passed on to the next steps in the analysis chain.
- for SLSTR data, a skin-to-depth correction factor is applied Donlon et al. (2002).

II.5.2 Bias correction of input satellite data

Satellite data can be biased for several reasons, including atmospheric water vapour; atmospheric aerosol (dust); surface changes (e.g. extreme roughness); instrument calibration problems. These biases can lead to biases in the analysis if they are not treated in some way. OSTIA uses a bias correction system based on match-up statistics between satellite and reference measurements (which are assumed to be unbiased). The reference dataset is currently specified to be moored and drifting buoy in situ data and a high-quality satellite dataset. Prior to 8th April 2012, AATSR data was included in the reference dataset. After 17th January 2013, a high-quality subset of MetOp-A AVHRR data was used. From 20th September 2016, night-time VIIRS SSTs were used for the bias correction and since 17th November 2020, dual-view SLSTR SSTs were used in addition to the VIIRS data. For each satellite observation type to be calibrated:

- match-ups are calculated between each reference data point and the satellite dataset (valid on the same day) with a spatial radius of 25 km.
- A large-scale objective analysis is calculated for each satellite observation type using the matchups as pseudo-observations of the bias, and a background from the previous day's bias analysis. The horizontal correlation scales are set to be 700 km for this bias analysis.
- The bias analysis field is interpolated back to the satellite observation locations, and the bias is subtracted from the satellite observation.

The outcome of this process is a new version of the satellite data, which has been bias-corrected.

II.5.3 Creation of the L4 analysis and error estimate

The main SST analysis uses a persistence-based approach using the previous analysis field as a **background with a relaxation to climatology**. The background field $x^b_{i,k}$ at grid point i and time k is defined as

$$x^b_{i,k} = \lambda_{i,k}(x^a_{i,k-1} - x^c_{i,k-1}) + x^c_{i,k} \qquad (1)$$

where $\lambda_{i,k}$ is a scalar less than 1, $x^a_{i,k-1}$ is the previous analysis, and $x^c_{i,k}$ is a reference climatology valid for the same time of year as time k. For each grid point and at each analysis time, a relaxation time scale is derived in order to determine $\lambda_{i,k}$. For ice-free areas this time scale is 30 days. SSTs under ice are relaxed toward 271.35 K with a shorter time scale which varies from 17.5 days to ~5 days linearly

with ice concentration from 50% to 100%. A digital Gaussian filter with a half-width of 4.7 km is applied to the background field to remove small scale noise.

The background field calculated using equation (1) and the bias corrected measurements (described previously) are then used to produce an analysis using a multi-scale Optimal Interpolation (OI) type scheme. An iterative procedure is used to calculate the OI solution that is both efficient and flexible when processing large numbers of observations. The system was upgraded to use the NEMOVAR variational data assimilation scheme on 13 February 2018 (see Section VI below, change dated 13th February 2018 for information on the impact of this change).

The **background error covariance matrix** is parameterised into two components. Both components have length scales and spatially varying variances associated with them. These have changed over time:

- Prior to 17th January 2013, one component had spatial correlation scales specified as 10 km and the other had spatial correlation scales of 100 km.
- On 17th January 2013 the background error covariances were improved; both components of the background error variance fields now varied seasonally, and the spatial correlations length scales were anisotropic and vary latitudinally. The shorter (mesoscale) length scales vary from 15 km to 40 km while the longer (synoptic) length scales vary from 200 km to 450 km.
- On 13th February 2018, following the upgrade to use the NEMOVAR scheme, the length scales were set to 0.36° and 2.70° globally, respectively.
- On March 12th 2019, the shorter length scale was changed to 0.135°.

The **observation error covariance matrix** is assumed to be a diagonal matrix (observation errors are uncorrelated with each other). The diagonal elements are specified using the SSES standard deviation values supplied with the GHRSST data. For in-situ data the diagonal elements are spatially varying observation errors that are specified a priori.

The **observation operator** is used to transform from the analysis grid to observation space. A number of different observation operators have been developed for use in OSTIA in order to represent the full range of satellite observation footprints. In the case of microwave data, for instance, the observation footprint is larger than the model grid, and the background gridded values which fall within the observation footprint are used to estimate the model equivalent of the observation. Following the upgrade to use NEMOVAR, the operator for all observations is an interpolation.

Each SST analysis value is accompanied by an **uncertainty estimate, which is sometimes referred to as an analysis error**. Various methods of approximating analysis uncertainty exist. The OSTIA system uses an analysis quality (AQ) approach to produce this estimate. In this scheme, a second analysis is performed that is identical to the main SST analysis except that all observations are given a value of 1.0, the background field is set to zero. The error estimates used in the main analysis (background & observational) are preserved. This field is then combined with the background error variance estimates described above to produce an analysis error estimate at each grid point.

II.6 Graphical examples of the L4 product

Examples of SST, sea-ice concentration and uncertainty fields are shown in Figure 2.

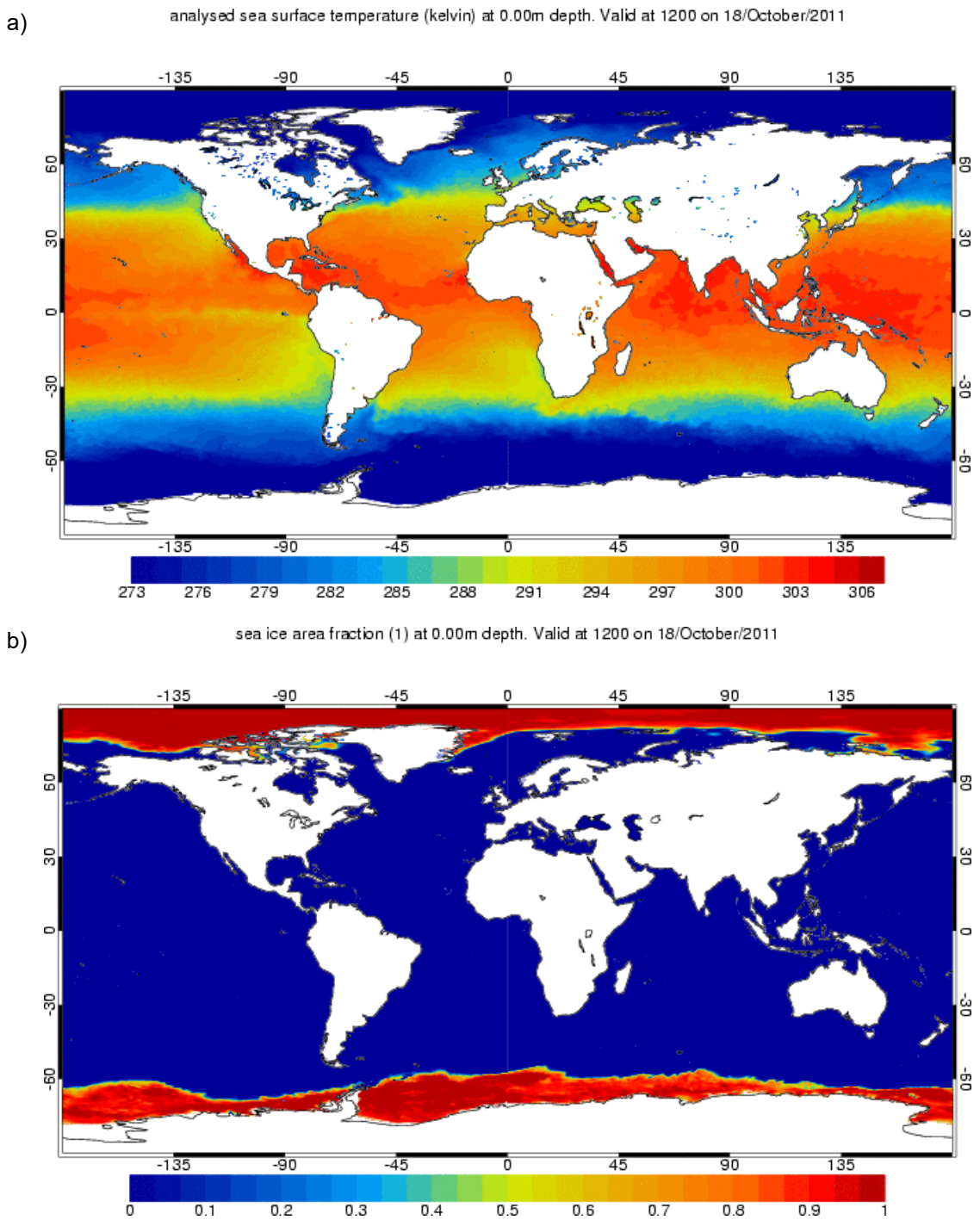

Figure 2. Examples of outputs from the high resolution daily L4 OSTIA product on 18th October 2011 (from top to bottom: (a) analysed SST, (b) sea-ice concentration).

III VALIDATION FRAMEWORK

An independent data source should be used to assess the true accuracy of an analysis. However, to deliberately withhold available data from an analysis can impact on the accuracy of the final product. This makes the validation of an SST reanalysis very challenging. Therefore, in this validation study, two assessments of the accuracy of the SST analysis are made. The first is to a subset of available drifting buoys which are not independent (because they are assimilated into the analysis) but have a good spatial coverage and have been used to validate other SST-TAC products (thus allowing easy comparison of accuracy between products). The second is to use near-surface Argo data which are independent of the analysis (i.e. Argo data are not assimilated) but offer a reduced number of observations relative to drifting buoys.

The analysis product includes an uncertainty estimate which is also validated here. The validation compares the uncertainty values to the differences between near surface Argo data and the analysis.

III.1 Drifting buoy comparison

This validation study uses drifting buoy measurements provided by the CMEMS (formerly MyOcean) IN-situ TAC for the year 2012. These measurements are a subset of the in-situ measurements that are used in the OSTIA processing. The analysis minus observation differences is calculated by bi-linearly interpolating the analysis to the observation locations for that day. The mean and standard deviation of these differences are then averaged globally and over various regions (as shown in Figure 3).

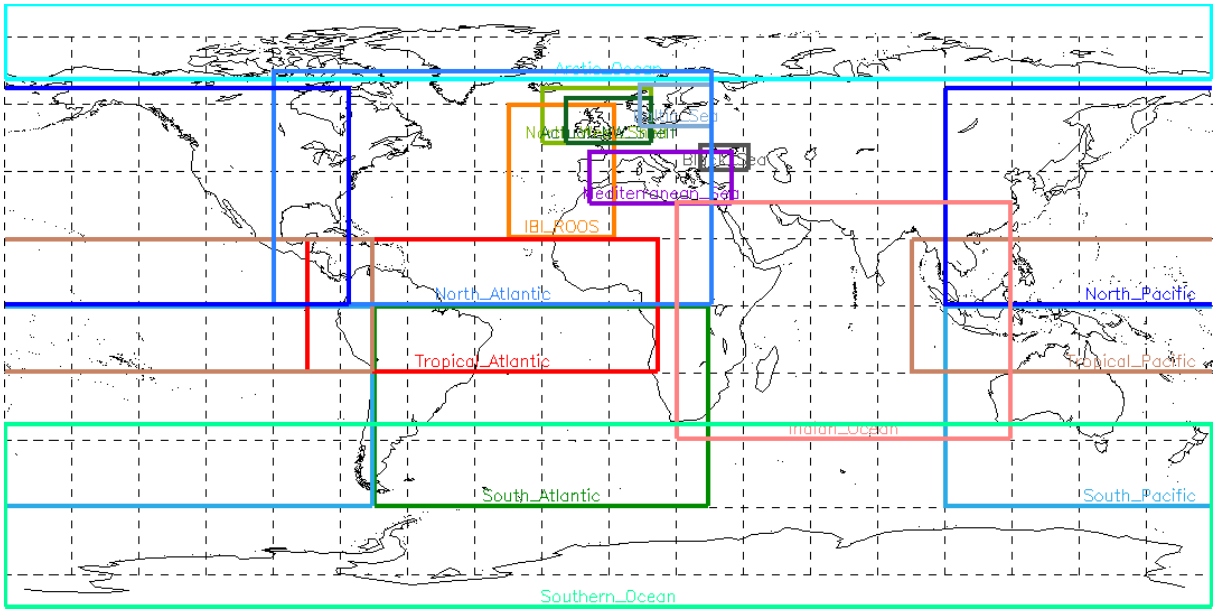

Figure 3. Extent of the geographical regions considered during the validation process.

III.2 Near-surface Argo comparison

Argo data have a high level of accuracy with an uncertainty value of 0.005 K (Oka & Ando, 2004) and although the Argo data coverage is less than drifting buoys, Argo floats now cover the majority of the world's oceans. Argo data has also been shown to agree well with drifting buoy data (e.g., Figure 1 in Fiedler et al., 2019). Near-surface Argo data has also previously been used to validate SST products (e.g. Martin et al., 2012; Corlett et al., 2014; Fiedler et al., 2014).

The accuracy of the analysis is assessed by comparison to Argo data using the shallowest available observations between 3-5m depth for the year 2013. The Argo data used here were obtained from the EN4 dataset (Good [2]et al., 2013). The analysis was bi-linearly interpolated to the Argo data locations. The mean and standard deviation of the Argo minus analysis differences were then calculated for the whole globe and over various ocean regions (as defined in Figure 3).

III.3 Analysis error validation

The standard deviation of the Argo minus analysis differences described above in section III.2 is also used to validate the analysis error estimate. The analysis error estimate was interpolated to the Argo data locations. Comparisons were made with the standard deviations for the whole globe and over various ocean regions. The analysis error estimates were also binned based on the magnitude of the analysis error and the standard deviation of the matchups corresponding to these estimates was found. Data between 5 February 2014 and 30 November 2014 were used in order to validate the newest implementation of the analysis error estimate calculation.

[2] www.metoffice.gov.uk/hadobs/en4

IV VALIDATION RESULTS

IV.1 Drifting buoy comparison

Table 4 details the mean and standard deviation of the analysis minus drifting buoy differences for the whole globe and for individual ocean regions. The mean difference is less than 0.1 K in all regions except the Mediterranean and Arctic regions, which suffer from a lack of available observations. The standard deviation varies between regions with the lowest value (suggesting highest accuracy) in the South Pacific and higher values in the North Atlantic and the Mediterranean.

Table 4. Global and regional mean (OSTIA minus observations) and standard deviation differences between OSTIA and drifting buoy data (from the CMEMS -formerly MyOcean- IN-Situ TAC) for 1 Jan 2012 to 31 Dec 2012.

Region	Mean difference (K)	Standard deviation of differences (K)	Number of observations
Global	0.06	0.40	7886293
Mediterranean	0.11	0.89	153646
North Atlantic	0.06	0.47	2290814
Tropical Atlantic	0.06	0.40	443619
South Atlantic	0.06	0.36	994086
North Pacific	0.04	0.36	1581671
Tropical Pacific	0.06	0.35	1162456
South Pacific	0.06	0.28	1669106
Indian Ocean	0.07	0.45	684185
Southern Ocean	0.06	0.33	1342172
Arctic	0.17	0.40	153676
North West Shelf	0.08	0.32	132256

IV.2 Near-surface Argo comparison

This section describes the original validation results performed for this product. Please see Section VI for updated results that were obtained as the product evolved over time.

Table 5 details the mean and standard deviation of the analysis minus Argo differences for the whole globe and for individual ocean regions. Note that the Arctic and Northwest Shelf regions are not included in Table 5 due to the lack of Argo observations in these regions. The results for the Mediterranean should also be treated with caution due to the low number of observations in the comparison.

The global statistics are comparable to the drifting buoy comparison with a small mean difference and a standard deviation of less than 0.5 K. The regional breakdown shows that some regions are more accurate than others, with South Atlantic being the region with the poorest accuracy in this comparison.

Table 5 . Global and regional mean (analysis minus observation) and standard deviation of differences between the OSTIA and near surface Argo data (from the EN4 dataset) for 1 Jan 2013 to 31 Dec 2013.

Region	Mean difference (K)	Standard deviation of differences (K)	Total number of observations
Global	-0.06	0.46	87947
Mediterranean Sea	0.01	0.35	1768
North Atlantic	-0.09	0.45	15299
Tropical Atlantic	-0.12	0.33	6155
South Atlantic	-0.11	0.59	7797
North Pacific	-0.05	0.50	27329
Tropical Pacific	0.00	0.29	23771
South Pacific	-0.03	0.37	22581
Indian Ocean	-0.02	0.36	14960
Southern Ocean	-0.17	0.55	16931

IV.3 Analysis error validation

Table 6 details the standard deviation of the analysis minus Argo differences for the whole globe and for individual ocean regions for 5 February to 30 November 2014. Also shown is the mean OSTIA analysis error estimate at the observation locations. These dates were chosen in order to validate the most recent data after a correction to the analysis error estimate calculation was implemented. Note that the Arctic and North West Shelf regions are not included in Table 6 due to the lack of Argo observations in these regions. The results for the Mediterranean should also be treated with caution due to the low number of observations in the comparison. The results demonstrate that globally and in all regions the OSTIA analysis error is underestimated compared to the Argo matchup standard deviations.

Table 6. Global and regional standard deviation of differences between OSTIA and near-surface Argo data (from the EN4 dataset) with mean OSTIA analysis error at Argo locations, for 5 Feb 2014 to 30 Nov 2014.

Region	Argo – OSTIA standard deviation (K)	Mean OSTIA analysis error (K)	Difference (K)	Total number of observations
Global	0.472	0.260	0.212	79550
Mediterranean Sea	0.419	0.308	0.111	1844
North Atlantic	0.535	0.279	0.256	14405
Tropical Atlantic	0.431	0.220	0.211	6557
South Atlantic	0.563	0.282	0.281	6498
North Pacific	0.463	0.271	0.192	28437
Tropical Pacific	0.305	0.214	0.091	21807
South Pacific	0.384	0.222	0.162	20178
Indian Ocean	0.376	0.235	0.141	9538
Southern Ocean	0.531	0.306	0.225	15321

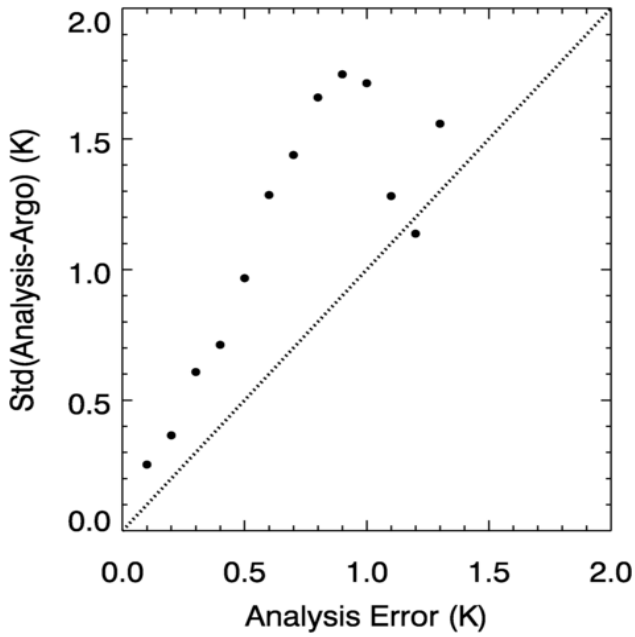

Figure 4. Standard deviation of OSTIA and Argo SST differences (y-axis) with OSTIA analysis error (x-axis) at Argo locations in 0.1 K bins, during 5 Feb 2014 to 30 Nov 2014. Dashed line is 1:1 relationship and assumes error in Argo observations is 0.0 K or negligible.

For Figure 4, the analysis error data is binned and compared with the standard deviation of matchups at the locations of the data within the bins. Figure 4 indicates that there is some correspondence between the analysis error and the Argo matchup differences with the analysis error generally increasing with increasing Argo difference. However, the analysis error is underestimated compared to Argo matchups at all magnitudes of the analysis error.

It should be noted that this finding is contrary to the findings of the ESA SST Climate Change Initiative (CCI) project which showed that the error estimates from the L4 product (which was generated with a reanalysis version of the OSTIA system) agreed well with drifting buoy matchup differences (Corlett et al., 2014). However, the SST CCI assessment compared the *robust* standard deviation of the drifting buoy matchup differences rather than the standard deviation as used in the above plot. It was possible to use drifting buoy data in SST CCI comparison as this data source was not used in generating the SST CCI L4 product and as such was an independent data source. The analysis error estimates in the near-real-time product and the SST CCI analysis product are expected to be very similar. Therefore, further work is required to examine the difference in the methodologies for assessing the error estimates before a firm conclusion on the accuracy of the analysis error estimates can be provided.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. In includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Date	Change/Event description	System version	Other
17 Jan 2013	Improved background error covariances and change to use MetOpB as reference sensor	2	
20 Sep 2016	Change to use SNPP VIIRS as reference sensor	2	
13 Feb 2018	System changed to use the NEMOVAR data assimilation scheme	3.0	
12 Mar 2019	Feature resolution of the analyses improved	3.2	
17 Nov 2020	SLSTR data added to the reference dataset	3.4	
29 Nov 2022	MetOp-C AVHRR included in the data used to generate the product	3.6	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

OSTIA system changes (post-2012) which have had a significant impact on accuracy or the product contents:

8th April 2012:

Loss of AATSR [3]on board ENVISAT, which had previously been used in the analysis and as a reference dataset to bias correct other input satellite data. In a test for March 2012, withholding AATSR data caused the global standard deviation of Argo –OSTIA differences to increase from 0.47 K to 0.53 K.

17th January 2013:

A number of improvements were made to improve the accuracy of the product and the feature resolution (i.e. the ability of the analyses to resolve SST features and gradients). The changes were:

- Update of the background error covariances in the OSTIA system:
    - New updated seasonal background error variances were used for both the mesoscale and synoptic scale components of the background errors.
    - Latitudinally varying anisotropic correlation length scales were used.
    - For in situ observations, type dependant measurement errors had been implemented which were added to spatially varying representativity errors.
    - The number of iterations of the assimilation had been increased from 10 to 100 iterations to ensure convergence.
    - The change increased the accuracy of the OSTIA SST analysis globally and regionally with respect to both assimilated (drifter and AATSR) data and with respect to independent ARGO observations. The table below (Table 7) presents global RMSD (in K), except for ARGO statistics which are standard deviation (in K), for March 2012. (The bias values are shown in brackets.) The feature resolution of the SST analysis had also improved due to the updates to error covariance parameters.

Table 7: Global statistics for comparison to drifters, AATSR and Argo for March 2012. Including root mean square differences (in K) for drifters, AATSR (bias in brackets) and standard deviation for ARGO (bias in brackets).

Validation type	OSTIA Control	OSTIA Updated
	RMSD (bias)	
Drifter o-b	0.52 (-0.01)	0.37 (0.00)
AATSR o-b	0.45 (0.04)	0.37 (0.03)
	Standard deviation (bias)	
ARGO o-a	0.47 (0.03)	0.43 (0.04)

[3] Advanced Along Track Scanning Radiometer

- The new estimates resulted in longer correlation length scales in the SST analysis. This caused information from SST observations to propagate further under the sea-ice and affects the consistency between the SST and sea-ice fields. However, overall, the new background error estimates improved the OSTIA SST analysis in the Arctic (with the drifter o-b RMSE decreasing from 0.58 K to 0.45 K).

- The reference data set used for the bias correction of the satellite data was expanded to include a subset of the most accurate MetOp AVHRR SST data (based on low satellite zenith angle). This replaced AATSR data which was included in the reference data set prior to its loss on 8th April 2012. The use of a subset of MetOp AVHRR data together with the in-situ data as a reference dataset increased the accuracy of the OSTIA SST analysis globally and regionally with respect to independent ARGO observations; the global standard deviation for March 2012 decreased from 0.53 K to 0.50 K.
- Change in the minimum quality flag of the geostationary satellite SST data used in the OSTIA analysis (SEVIRI/MSG data; GOES-E data). This change was negligible in terms of the global and regional RMSE and mean bias validation statistics of the OSTIA analysis. Using data flagged as lower quality increased the data coverage of the geostationary satellite observations which improved the SST locally. Of particular relevance to European users was the improved SEVIRI coverage on the Northwest European Shelf.
- Set a minimum SST on the OSTIA SST analysis. This change would have negligible impact in terms of the global and regional RMSE and mean bias validation statistics of the OSTIA analysis. It would however stop negative increments pushing the analysed SST below a minimum temperature (set as -2°C).

17th April 2013:

Lake ice was added to OSTIA ice field using NCEP data for ice in inland waters. This change included a slight relaxation to 0ºC under lake ice but otherwise there was no effect on global ocean SST.

11th November 2013:

A change was made to the OSTIA system to routinely produce monthly and seasonal mean files of the L4 product.

5th February 2014:

A change was made to OSTIA to correct the computation of the analysis error estimates in the system. This fixed a problem in the system that was present during the period 17 Jan 2013 to 4 Feb 2014 and would have caused the analysis error to be overestimated during this time. The SST analysis and sea ice fields were unaffected by this correction.

23rd February 2016:

The high-quality subset of MetOp-A AVHRR observations used in the bias correction reference dataset was updated to MetOp-B AVHRR due to an upstream change in data availability.

15th March 2016: Two new global SST datasets were included in the OSTIA analysis: an infra-red dataset from the VIIRS instrument and a microwave dataset from the AMSR2 instrument. The inclusion of these data sets had led to an improvement in the OSTIA analysis accuracy. Table 8. below shows a comparison between accuracy statistics for the old and updated analyses for January 2016, using near surface Argo observations (from the EN4 database) as a reference.

20th September 2016: Nighttime ACSPO VIIRS SST product was used in OSTIA for as the satellite reference for the bias correction, replacing MetOp-A AVHRR data. From the test run over the period of 09 December 2015 – 11 January 2016, this change had led to an improvement if the OSTIA analysis accuracy. Table 9. shows a comparison between the accuracy statistics for the old (Control) and updated (VIIRSG_ref) analyses for the test period, using near surface Argo observations (from the EN4 database) as a reference.

13th February 2018:

The data assimilation scheme used to generate the product was changed to the NEMOVAR variational scheme. The sea ice field included in the files began to be generated using NEMOVAR rather than a regridding of the input observations. The impact on accuracy statistics is summarised in Table 10. Generally the new system was found to perform better in these statistics than the old version.

12th March 2019

The feature resolution of the analyses was improved. Users would see larger gradients in the SST fields and sharper features. This is illustrated in Figure 5 below. The set of data sources used to generate the SST analysis was expanded to include SSTs from the Sea and Land Surface Temperature Radiometer (SLSTR) on Sentinel 3A. The impact on the accuracy statistics is summarised in Table 11 (for winter 2017/18) and Table 12 (summer 2018). Improvements of 0.02 K or more are highlighted in bold. The major impact was a reduction in the standard deviation of differences between Argo reference data and the analyses. Improvements of 0.02 K or more occurred in both seasons for the North Atlantic, South Atlantic and the Southern Ocean, in winter only globally and in the North Pacific, and in summer only for the South Pacific and Indian Ocean. Smaller differences (of 0.01) were also seen with standard deviations of differences tending to be smaller after the change but mean differences larger.

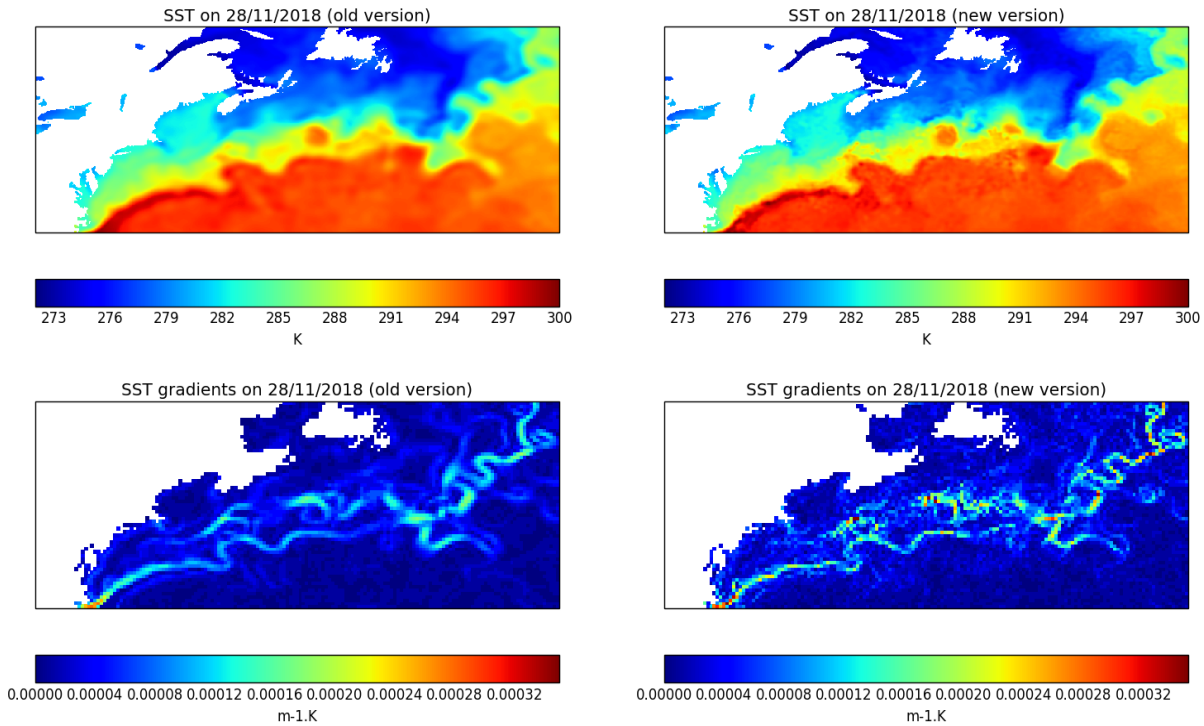

Figure 5. Illustration of the change in sharpness in features and increase in the strength of SST gradients due to the 12th March 2019 upgrade.

12th November 2019

AVHRR from the NOAA19 satellite was no longer used in the generation of the product but SLSTR from Sentinel 3B and VIIRS from NOAA20 were now used. The impact on the quality of the product was neutral.

Table 8. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) and the old and updated OSTIA analyses for January 2016.

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	Old analysis	with VIIRS and AMSR2	Old analysis	with VIIRS and AMSR2	
Global	0.12	0.11	0.49	0.42	9580
North Atlantic	0.23	0.21	0.60	0.47	1720
Tropical Atlantic	0.14	0.14	0.27	0.25	837
South Atlantic	0.02	0.03	0.60	0.50	727
North Pacific	0.20	0.17	0.47	0.41	2973
Tropical Pacific	0.08	0.06	0.32	0.27	2479
South Pacific	0.03	0.03	0.40	0.35	2696
Indian Ocean	0.04	0.05	0.34	0.30	1316
Southern Ocean	0.07	0.06	0.52	0.45	2172

Table 9. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) and the old and updated OSTIA analyses for 09 December 2015 – 11 January 2016

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	Control	VIIRSG_ref	Control	VIIRSG_ref	
Global	0.12	0.06	0.44	0.40	10432
North Atlantic	0.22	0.05	0.46	0.45	1929
Tropical Atlantic	0.17	0.11	0.23	0.21	899
South Atlantic	0.08	0.08	0.48	0.45	798
North Pacific	0.20	0.09	0.49	0.46	3384
Tropical Pacific	0.08	0.07	0.25	0.21	2707
South Pacific	0.03	0.07	0.32	0.29	2796
Indian Ocean	0.03	0.09	0.29	0.27	1296
Southern Ocean	0.07	0.04	0.45	0.43	2182

Table 10. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) and the OSTIA analyses produced using an OI-like scheme and the NEMOVAR OSTIA analyses for 1 July 2016 to 31 March 2017.

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	OI OSTIA	NEMOVAR OSTIA	OI OSTIA	NEMOVAR OSTIA	
Global	0.06	0.06	0.41	0.39	90171
North Atlantic	0.06	0.06	0.51	0.49	16583
Tropical Atlantic	0.13	0.12	0.43	0.43	6322
South Atlantic	0.09	0.09	0.50	0.48	7240
North Pacific	0.03	0.03	0.40	0.38	27283
Tropical Pacific	0.04	0.04	0.29	0.27	23182
South Pacific	0.07	0.07	0.32	0.31	24262
Indian Ocean	0.10	0.10	0.32	0.30	13024
Southern Ocean	0.07	0.08	0.41	0.40	19484

Table 11. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) showing the impact of the 12th March 2019 upgrade (upgr.), for the period December 2017 to February 2018. Where statistics are different between versions by 0.02 or more, the better number is highlighted in bold.

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	Without 12th March upgr.	With 12th March upgr.	Without 12th March upgr.	With 12th March upgr.	
Global	0.03	0.04	0.36	**0.34**	29981
North Atlantic	0.03	0.04	0.53	**0.49**	4313
Tropical Atlantic	0.14	0.13	0.29	0.28	1731
South Atlantic	0.03	0.04	0.37	**0.35**	2972
North Pacific	0.04	0.05	0.35	**0.33**	7569
Tropical Pacific	0.03	0.03	0.24	0.24	6694
South Pacific	0.04	0.04	0.30	0.29	9239
Indian Ocean	0.05	0.05	0.28	0.27	4994
Southern Ocean	0.02	0.03	0.38	**0.36**	7883

Table 12. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) showing the impact of the 12th March 2019 upgrade, for the period June 2018 to August 2018. Where statistics are different between versions by 0.02 or more, the better number is highlighted in bold.

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	Without 12th March upgr.	With 12th March upgr.	Without 12th March upgr.	With 12th March upgr.	
Global	0.04	0.05	0.40	0.39	30291
North Atlantic	0.01	0.02	0.54	**0.52**	5340
Tropical Atlantic	0.10	0.10	0.30	0.30	2280
South Atlantic	0.04	0.05	0.46	**0.42**	3495
North Pacific	0.03	0.04	0.41	0.41	7292
Tropical Pacific	0.03	0.03	0.22	0.22	7029
South Pacific	0.03	0.03	0.29	**0.27**	9543
Indian Ocean	0.12	0.12	0.40	**0.38**	4730
Southern Ocean	0.04	0.05	0.41	**0.38**	7166

17th November 2020

The reference dataset used for calculating the bias in data from satellite sensors not included in the reference was changed. Dual-view SLSTR data were now included in addition to night-time VIIRS and drifting and moored buoy in situ data. The impact was to reduce the bias in the analysis according to comparisons with Argo data, as shown in Table 13.

Table 13. Global and regional mean (analysis minus observation) and standard deviation of differences between near surface Argo data (from the EN4 dataset) showing the impact of the 17th November 2020 upgrade, for June 2020. Where statistics are different between versions by 0.02 or more, the better number is highlighted in bold.

Region	Mean difference (K)		Standard deviation of differences (K)		Total number of Argo observations
	Without 12th March upgr.	With 12th March upgr.	Without 12th March upgr.	With 12th March upgr.	
Global	-0.08	**-0.04**	0.34	0.34	10227
North Atlantic	-0.04	0.04	0.47	0.48	2101
Tropical Atlantic	-0.14	**-0.08**	0.26	**0.24**	777
South Atlantic	-0.08	**-0.06**	0.32	0.31	1126
North Pacific	-0.1	**-0.04**	0.31	0.31	2274
Tropical Pacific	-0.07	**-0.03**	0.18	0.18	2134
South Pacific	-0.08	**-0.06**	0.24	0.23	3313
Indian Ocean	-0.14	**-0.09**	0.3	**0.28**	1289
Southern Ocean	-0.1	-0.09	0.34	0.34	2500

29th November 2019

AVHRR data from MetOp-C were included in the production of the product. This caused minimal changes in quality statistics but added robustness to the system.

24th June 2025

Replace SNPP-VIIRS with NOAA21-VIIRS due to the aging of the satellite. The NEMOVAR used in the processing is updated to a newer version. These changes are expected to have a neutral impact on the system.

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